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Lower Oxygen Tension and Intracranial Hemorrhage in Venovenous Extracorporeal Membrane Oxygenation

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Abstract

We examined the relationship between 24-hour pre- and post-cannulation arterial oxygen tension (PaO₂) and arterial carbon dioxide tension (PaCO₂) and subsequent acute brain injury (ABI) in patients receiving venovenous extracorporeal membrane oxygenation (VV-ECMO) with granular arterial blood gas (ABG) data and institutional standardized neuromonitoring. Eighty-nine patients underwent VV-ECMO (median age=50, 63% male). Twenty (22%) patients experienced ABI; intracranial hemorrhage (ICH) was the most common diagnosis (n=14, 16%). Lower post-cannulation PaO₂ levels were significantly associated with ICH (66 vs. 81 mmHg, p=0.007) and a post-cannulation PaO₂ level <70 mmHg was more frequent in these patients (71% vs. 33%, p=0.007). PaCO₂ parameters were not associated with ABI. By multivariable logistic regression, hypoxemia post-cannulation increased the odds of ICH (OR=5.06, 95% CI:1.41–18.17; p=0.01). In summary, lower oxygen tension in the 24-hours post-cannulation was associated with ICH development. The precise roles of peri-cannulation ABG changes deserve further investigation, as they may influence the management of VV-ECMO patients.

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Author Contributions

AFA, BLS, and SMC contributed to the study conception and design. Material preparation and data collection were performed by AFA, BLS, and DXZ. Data analysis was performed by AFA. The first draft of the manuscript was written by AFA and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics Approval

This was an observational study. This study was approved by the Johns Hopkins Institutional Review Board with a waiver of informed consent.

Keywords

Extracorporeal Membrane Oxygenation; ECMO; Arterial Oxygen Tension; Arterial Carbon Dioxide Tension; Acute Brain Injury; Neurological Injury; Neurological Complication

Introduction

Acute brain injury (ABI) occurs in 5–13%^{1–3} of patients undergoing veno-venous extracorporeal membrane oxygenation (VV-ECMO). Arterial carbon dioxide tension (PaCO₂) is a potent cerebral vasodilator, increasing cerebral blood flow (CBF)⁴ and neuronal metabolic demand⁵. Hypercapnia causes the plateau phase of the cerebral autoregulation curve to shift upward and shorten, leading to persistently increased CBF and a narrower regulatory pressure window⁶. Thus, rapid reduction of hypercapnia has mechanistic plausibility to cause ABI. Arterial oxygen tension (PaO₂) also affects autoregulation, and due to the brain's high metabolic demand, acute hypoxia causes cerebral vasodilation and increases CBF through direct effects on cerebral vasculature⁷. Elevated CBF has been shown with PaO₂ values below 65 mmHg⁸ and arterial oxygen saturation (SpO₂) values below 90%⁹. Hypoxia increases local nitric oxide and adenosine production and opens ATP-dependent potassium channels on vasculature, collectively contributing to vasodilation^{10–12}.

Cavayas et al. reported an association between early PaCO₂ changes and intracranial hemorrhage (ICH) using multi-center ELSO data in VV-ECMO². A single-center retrospective study corroborated this and additionally reported an association between large PaO₂ increases after ECMO initiation and ICH³. However, these studies have limited arterial blood gas (ABG) data points and lack standardized neuromonitoring protocol. Using our institution's granular ABG data and standardized neuromonitoring protocol, we aimed to describe the relationship between early PaCO₂ and PaO₂ changes and ABI in VV-ECMO. We hypothesized that post-cannulation PaCO₂ drops are associated with ICH.

Methods

We retrospectively analyzed all VV-ECMO patients (≥18 years old) at a tertiary care center between September 2016 and March 2022. ABGs were obtained every 2–4 hours. Patients with significant neurological issues and pre-existing recent brain injury were excluded. Patients underwent neurocritical care consultations with standardized neuromonitoring¹³, which involves neurological examinations, transcranial Doppler, electroencephalography, and somatosensory evoked potentials at pre-specified intervals from ECMO day 1 until decannulation. Pre- and post-cannulation ABG values were defined as the median of all PaCO₂ and PaO₂ values, respectively, 24 hours before and after cannulation. Δ PaCO₂ equaled maximum pre-cannulation minus minimum post-cannulation PaCO₂. Hypoxemia post-cannulation was defined as a median post-cannulation PaO₂ value <70 mmHg¹⁴. We also examined duration of post-cannulation hypoxemia, defined as SpO₂ <88%. Favorable and unfavorable neurological outcome were defined as a modified Rankin Score (mRS) ≤3 and >3 at discharge, respectively. Primary outcome was presence of composite ABI, including ICH, ischemic stroke, hypoxic ischemic brain injury, cerebral edema, seizure, and brain death. Secondary outcomes were individual ABI diagnoses and in-hospital mortality.

Descriptive statistics was used to compare associations between ABI diagnoses and ABG parameters.

Results

Eighty-nine patients (63% male) underwent VV-ECMO (median age=50, interquartile range [IQR]=39–58) (Supplementary Table 1). Median ECMO support duration was 439 hours (IQR=185–824). Forty-five patients (51%) died during hospitalization. Twenty (22%) patients experienced at least one ABI during ECMO support. ICH was most common (n=14, 16%), followed by ischemic stroke (n=5, 6%), seizure (n=4, 4%), and brain death (n=4, 4%). Those with ABI (vs. without) required more pre-ECMO vasoactive support (60% vs. 33%, $p=0.032$) and trended toward higher Sequential Organ Failure Assessment (SOFA) scores (11 vs. 9, $p=0.069$). Those without ABI and without ICH were more frequently hypertensive (46% vs. 20%, $p=0.034$; 45% vs. 14%, $p=0.030$). Patients with ICH were less frequently discharged home (0% vs. 29%).

A median of 4 (2–7) pre-cannulation and 5 (3–8) post-cannulation ABG values were recorded per patient. All PaCO₂ values 24 hours before and after cannulation are plotted in Figure 1a. Median PaCO₂ was significantly higher before cannulation compared to after (60 vs. 44 mmHg, $p<0.001$), while median PaO₂ was not significantly different (76 vs. 79 mmHg, $p=0.26$). Median Δ PaCO₂ was 33 (25–45) mmHg. Thirty patients (34%) experienced hypoxemia (PaO₂<70 mmHg) before ECMO, thirty-five (39%) after cannulation.

In those with ABI and ICH versus those without, there was no significant difference in pre-cannulation PaCO₂, post-cannulation PaCO₂ or Δ PaCO₂ (Table 1). However, patients with ABI tended to be more frequently hypoxemic (55% vs. 35%, $p=0.10$) and have lower post-cannulation PaO₂ (68 vs. 79 mmHg, $p=0.12$) than those without. Those with ICH (n=14, 16%) were significantly more frequently hypoxemic (71% vs. 33%, $p=0.007$) with lower post-cannulation PaO₂ (66 vs. 81 mmHg, $p=0.007$) than those without. Figure 1b shows that while lower post-cannulation PaO₂ was associated with ICH, there was no association with post-cannulation PaCO₂.

By multivariable logistic regression controlling for post-cannulation PaCO₂, patients with hypoxemia post-cannulation were more likely to experience ICH (OR=5.06, 95% CI:1.41–18.17; $p=0.01$) (Table 2). Decreased minimum post-cannulation PaO₂ was significantly associated with a higher risk of ICH (OR=2.13, 95% CI:1.08–4.17; $p=0.03$), as was decreased median post-cannulation PaO₂ (OR=1.61, 95% CI:1.04–2.50; $p=0.03$). Neither pre-cannulation PaCO₂ nor Δ PaCO₂ were associated with increased odds of ICH when controlling for post-cannulation PaO₂. Figure 1c shows that as minimum PaO₂ decreases post-cannulation, ICH probability increases.

Additional analyses examined hypoxemia duration (SpO₂<88%) in cannulated patients up to 30 days post-cannulation (Supplementary Table 2). Higher duration of hypoxemia was associated with increased mortality (1343 vs. 495 minutes, $p=0.02$) and unfavorable neurological outcome (1200 vs. 495 minutes, $p=0.04$).

Discussion

This study found that lower oxygen tension after VV-ECMO cannulation was associated with the development of ICH. Cavayas et al. demonstrated an association between pericannulation PaCO₂ decreases and ICH using multi-center retrospective ELSO data, but to our knowledge, this is the first study to report an association between post-cannulation hypoxemia and ICH². Muellenbach et al. reported that at VV-ECMO initiation, patients were at risk of reduction in regional cerebral tissue oxygen saturation, and linked hypoxia to cerebral hypoperfusion due to carbon dioxide elimination¹⁵. Our patients uniformly experienced large peri-cannulation PaCO₂ drops. This was not independently associated with ICH; however, PaCO₂-mediated vasoconstriction might have impaired cerebral oxygenation as posited by Muellenbach, leading to subsequent ABI.

Despite VV-ECMO cannulation, refractory hypoxemia was common in our cohort. Due to the brain's high metabolic demand, acute hypoxia can cause cerebral vasodilation and increase CBF through direct and indirect effects cerebral vasculature^{7,10–12}. This impairs pressure-directed cerebral autoregulation. Acute drops in PaCO₂ shift the oxygen-hemoglobin dissociation curve to the left, which further impedes oxygen delivery to brain tissue. In the setting of VV-ECMO with refractory hypoxemia, as well as impaired tissue oxygen delivery, there is a mechanistic plausibility for hypoxic-ischemic insults and subsequent intracranial hemorrhage. Additionally, when ECMO is initiated, patients are anticoagulated with a heparin bolus, which increases bleeding risk. Blood contacts the circuit, activating inflammatory and coagulation pathways that can drive thrombotic or hemorrhagic sequelae. While our data suggest that lower oxygen tension may play a role, the etiology of ICH is likely multifactorial due to a variety of potential insults that may affect the susceptible brain in the early ECMO period.

Despite our cohort's large peri-cannulation Δ PaCO₂ of 33 (25–45) mmHg, we found no association between Δ PaCO₂ and ICH, in contrast to previous studies of ECMO patients^{16,17}. In an ELSO study of mostly VV-ECMO patients, those with relative PaCO₂ decreases of >50% between pre- and post-ECMO initiation were more likely to experience composite ABI and ICH¹⁷. Luyt et al. also found that large decreases (>27 mmHg) in PaCO₂ were associated with ICH in a single-center VV-ECMO study¹⁶. Both studies recorded only one pre- and post-ECMO ABG value, in contrast to our granular ABG data, and may fail to capture the dynamic status of patients' oxygen state in the 24-hour peri-cannulation period. We recently reported that PaCO₂ drops, of a much smaller scale (8.8 vs. 4.5 mmHg), were associated with ICH in veno-arterial ECMO patients¹⁸.

Our study encourages maximizing systemic oxygenation in VV-ECMO; it may be necessary to consider novel strategies in patients who remain hypoxemic post-cannulation and potentially expand PaO₂ targets above 70 mmHg. While PaO₂ is considered low-normal, factors like concurrent PaCO₂, hemoglobin concentration and saturation, pH, and body temperature may play a role in oxygen delivery and susceptibility to brain injury¹⁹. Our retrospective, single-institution study may not reflect broader practice patterns, especially considering ECMO delivery varies by institution. We had a small population of 89 patients with only 14 instances of ICH, restricting our statistical power. Despite this, our institution's

standardized neuromonitoring protocol resulted in a high sensitivity of ABI detection, with 22% of patients experiencing ABI. In our prior study, this protocol increased detection of ABI and improved neurological outcomes at discharge, indicating it might prevent worsening of ABI with timely interventions²⁰. The threshold of 70 mmHg, below which we observed increased ICH, is not low enough to be considered hypoxemic in the literature. This threshold's clinical utility must be confirmed by larger studies.

Conclusions

Using granular ABG data and standardized neuromonitoring protocol, we demonstrated that lower oxygen tension in the early post-cannulation period within 24 hours was associated with ICH development in VV-ECMO patients, but arterial carbon dioxide change at cannulation was not. The precise roles of ABG changes in the peri-cannulation period need to be elucidated, as they may influence the clinical management of VV-ECMO patients.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

1. Cho SM, Canner J, Caturegli G, et al. Risk Factors of Ischemic and Hemorrhagic Strokes during Venovenous Extracorporeal Membrane Oxygenation: Analysis of Data from the Extracorporeal Life Support Organization Registry. *Crit Care Med*. Published online 2020:91–101. doi:10.1097/CCM.0000000000004707 [PubMed: 31725438]
2. Cavayas YA, Munshi L, Del Sorbo L, Fan E. The Early Change in PaCO₂ after Extracorporeal Membrane Oxygenation Initiation Is Associated with Neurological Complications. *Am J Respir Crit Care Med*. 2020;201(12):1525–1535. doi:10.1164/rccm.202001-0023OC [PubMed: 32251606]
3. Luyt CE, Bréchet N, Demondion P, et al. Brain injury during venovenous extracorporeal membrane oxygenation. *Intensive Care Med*. 2016;42(5):897–907. doi:10.1007/s00134-016-4318-3 [PubMed: 27007107]
4. HARPER AM, BELL RA. The effect of metabolic acidosis and alkalosis on the blood flow through the cerebral cortex. *J Neurol Neurosurg Psychiatry*. 1963;26(4):341–344. doi:10.1136/jnnp.26.4.341 [PubMed: 14043049]
5. Dulla CG, Dobelis P, Pearson T, Frenguelli BG, Staley KJ, Masino SA. Adenosine and ATP link PCO₂ to cortical excitability via pH. *Neuron*. 2005;48(6):1011–1023. doi:10.1016/j.neuron.2005.11.009 [PubMed: 16364904]
6. Meng L, Gelb AW. Regulation of cerebral autoregulation by carbon dioxide. *Anesthesiology*. 2015;122(1):196–205. doi:10.1097/ALN.0000000000000506 [PubMed: 25401418]
7. Masamoto K, Tanishita K. Oxygen Transport in Brain Tissue. *J Biomech Eng*. 2009;131(7). doi:10.1115/1.3184694
8. Ellingsen I, Hauge A, Nicolaysen G, Thoresen M, Walløe L. Changes in human cerebral blood flow due to step changes in PAO₂ and PACO₂. *Acta Physiol Scand*. 1987;129(2):157–163. doi:10.1111/j.1748-1716.1987.tb08054.x [PubMed: 3554898]

9. Gupta AK, Menon DK, Czosnyka M, Smielewski P, Jones JG. Thresholds for hypoxic cerebral vasodilation in volunteers. *Anesth Analg*. 1997;85(4):817–820. doi:10.1097/0000539-199710000-00018 [PubMed: 9322461]
10. Van Mil AHM, Spilt A, Van Buchem MA, et al. Nitric oxide mediates hypoxia-induced cerebral vasodilation in humans. *J Appl Physiol*. 2002;92(3):962–966. doi:10.1152/japplphysiol.00616.2001 [PubMed: 11842027]
11. Winn HR, Rubio GR, Berne RM. The Role of Adenosine in the Regulation of Cerebral Blood Flow. *J Cereb Blood Flow Metab*. 1981;1(3):239–244. doi:10.1038/jcbfm.1981.29 [PubMed: 7328142]
12. H T, Dd H, T K, Fm F. ATP-sensitive K⁺ channels mediate dilatation of cerebral arterioles during hypoxia. *Circ Res*. 1994;74(5). doi:10.1161/01.res.74.5.1005
13. Cho SM, Ziai W, Mayasi Y, et al. Noninvasive Neurological Monitoring in Extracorporeal Membrane Oxygenation. *ASAIO J*. Published online 2020:388–393. doi:10.1097/MAT.0000000000001013 [PubMed: 31045914]
14. Samuel J, Franklin C. Hypoxemia and Hypoxia. In: Myers JA, Millikan KW, Saclarides TJ, eds. *Common Surgical Diseases*. Springer New York; 2008:391–394. doi:10.1007/978-0-387-75246-4_97
15. Muellenbach RM, Kilgenstein C, Kranke P, et al. Effects of venovenous extracorporeal membrane oxygenation on cerebral oxygenation in hypercapnic ARDS. *Perfusion*. 2014;29(2):139–141. doi:10.1177/0267659113497073 [PubMed: 23887087]
16. Luyt CE, Bréchet N, Demondion P, et al. Brain injury during venovenous extracorporeal membrane oxygenation. *Intensive Care Med*. 2016;42(5):897–907. doi:10.1007/s00134-016-4318-3 [PubMed: 27007107]
17. Cavayas YA, Munshi L, Del Sorbo L, Fan E. The Early Change in PaCO₂ after Extracorporeal Membrane Oxygenation Initiation Is Associated with Neurological Complications. *Am J Respir Crit Care Med*. 2020;201(12):1525–1535. doi:10.1164/rccm.202001-0023OC [PubMed: 32251606]
18. Shou BL, Ong CS, Zhou AL, et al. Arterial Carbon Dioxide and Acute Brain Injury in Venovenous Extracorporeal Membrane Oxygenation. *ASAIO J Am Soc Artif Intern Organs* 1992. Published online March 16, 2022. doi:10.1097/MAT.0000000000001699
19. Ziaka M, Exadaktylos A. ARDS associated acute brain injury: from the lung to the brain. *Eur J Med Res*. 2022;27(1):150. doi:10.1186/s40001-022-00780-2 [PubMed: 35964069]
20. Ong CS, Etchill E, Dong J, et al. Neuromonitoring detects brain injury in patients receiving extracorporeal membrane oxygenation support. *J Thorac Cardiovasc Surg*. Published online October 30, 2021:S0022–5223(21)01508–7. doi:10.1016/j.jtcvs.2021.09.063

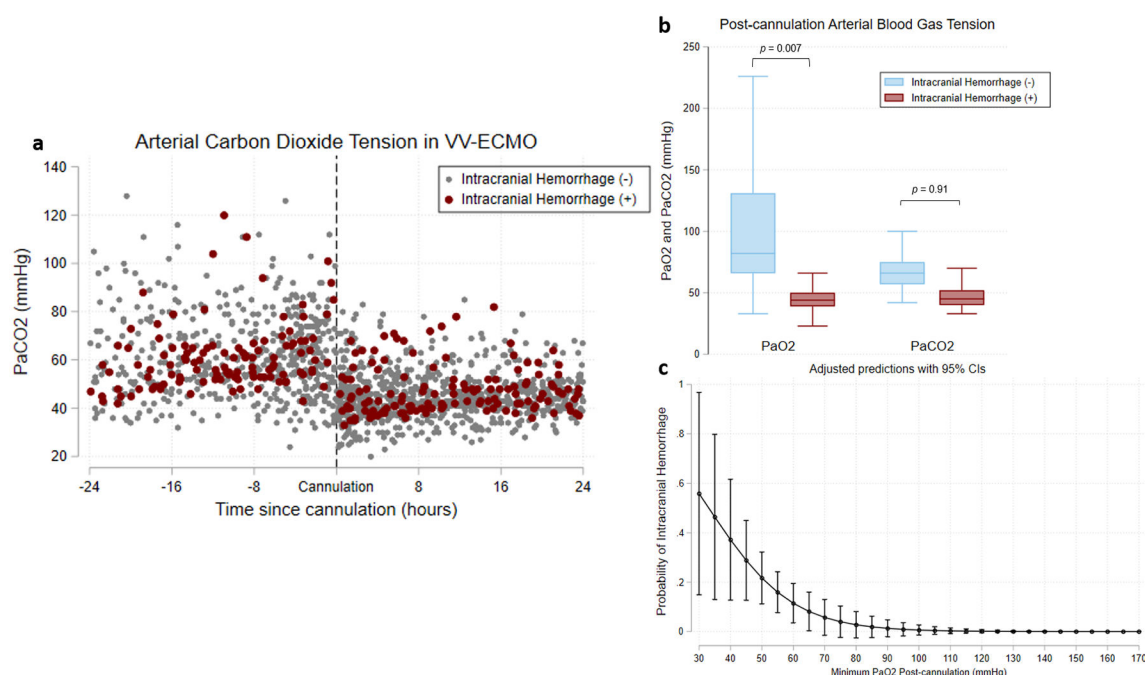


Figure 1.

Relationships between arterial blood gas tension and intracranial hemorrhage in VV-ECMO.

a: All collected arterial carbon dioxide values in the 24 hours before and after cannulation.

Red and gray dots represent PaCO₂ values for patients with and without intracranial hemorrhages, respectively. **b:** All 24-hour post-cannulation PaO₂ and PaCO₂ values are compared between those who experienced intracranial hemorrhage and those who did not. Post-cannulation PaO₂ values are lower in patients who experienced intracranial hemorrhage, while post-cannulation PaCO₂ values are not different. **c:** Probability of intracranial hemorrhage increases with decreasing minimum PaO₂ post-cannulation.

Table 1.**Arterial Blood Gas Tension and Incidence of Acute Brain Injury**

ABG Parameter	Patients without ABI (n = 69)	Patients with ABI (n = 20)	<i>p</i> -value	Patients without ICH (n = 75)	Patients with ICH (n = 14)	<i>p</i> -value
Peri-cannulation PaO ₂ parameters						
Pre-cannulation						
Median	76 (67–89)	73 (66–84)	0.44	77 (68–91)	69 (64–78)	0.13
Highest	101 (82–148)	108 (80–157)	0.89	103 (82–175)	98 (78–108)	0.17
Lowest	60 (51–69)	60 (55–67)	0.77	61 (51–69)	60 (54–63)	0.73
Post-cannulation						
Median	79 (68–113)	68 (63–87)	0.12	81 (68–114)	66 (61–80)	0.007
Highest	144 (97–249)	113 (80–213)	0.30	145 (93–271)	112 (74–157)	0.057
Lowest	58 (50–67)	54 (49–59)	0.17	58 (50–67)	50 (49–54)	0.008
ΔPaO ₂	46 (23–90)	51 (30–99)	0.44	48 (25–119)	44 (25–59)	0.49
Hypoxemia post-cannulation	24 (35%)	11 (55%)	0.10	25 (33%)	10 (71%)	0.007
Peri-cannulation PaCO ₂ parameters						
Pre-cannulation						
Median	61 (52–74)	59 (52–67)	0.69	60 (50–74)	60 (54–66)	0.79
Highest	74 (61–87)	68 (60–80)	0.22	73 (61–87)	68 (60–83)	0.64
Lowest	54 (43–65)	49 (44–60)	0.59	52 (42–65)	56 (45–61)	0.55
Post-cannulation						
Median	45 (40–49)	42 (40–47)	0.62	45 (40–49)	43 (41–48)	0.91
Highest	54 (50–63)	53 (50–58)	0.61	54 (50–63)	52 (48–59)	0.47
Lowest	39 (34–42)	38 (36–40)	0.79	38 (33–42)	39 (36–40)	0.55
ΔPaCO ₂	35 (26–46)	30 (18–44)	0.21	35 (25–46)	30 (18–45)	0.51

p values are from Wilcoxon rank-sum test. PaO₂ and PaCO₂ values are represented as median (IQR). All values are expressed as mm Hg. Bold entries indicate a *p* < 0.05. Pre- and post- cannulation PaO₂ and PaCO₂ values are calculated from the 24-hour period before and after cannulation, respectively. ΔPaO₂ and ΔPaCO₂ are calculated from the difference between the highest value pre-cannulation and the lowest value post-cannulation. Post-cannulation hypoxemia is defined as a median PaO₂ value below 70 mm Hg.

ΔPaO₂, peri-cannulation arterial oxygen drop; ΔPaCO₂, peri-cannulation arterial carbon dioxide drop; ABG, arterial blood gas; ABI, acute brain injury; ICH, intracranial hemorrhage; IQR, interquartile range; PaO₂, arterial oxygen tension; PaCO₂, arterial carbon dioxide tension

Table 2.

Multivariable Logistic Regression for Incidence of Intracranial Hemorrhage (ICH)

Variable	OR (95% CI)	<i>p</i>
Median Pre-cannulation PaO ₂ *	0.86 (0.67–1.11)	0.23
Minimum Pre-cannulation PaO ₂ *	1.01 (0.77–1.33)	0.94
Median Post-cannulation PaO ₂ *	0.62 (0.40–0.96)	0.03
Minimum Post-cannulation PaO ₂ *	0.47 (0.24–0.93)	0.03
ΔPaO ₂ *	0.93 (0.85–1.03)	0.15
Hypoxemia post-cannulation	5.06 (1.41–18.17)	0.01
Median Pre-cannulation PaCO ₂ †	0.94 (0.58–1.50)	0.78
Maximum Pre-cannulation PaCO ₂ †	0.86 (0.59–1.27)	0.45
Median Post-cannulation PaCO ₂ †	0.82 (0.36–1.88)	0.64
Minimum Post-cannulation PaCO ₂ †	0.66 (0.25–1.78)	0.41
ΔPaCO ₂ †	0.91 (0.62–1.35)	0.65

Positive values for ΔPaCO₂ and ΔPaO₂ represent drops (decreases) in PaCO₂. Bold entries indicate a *p* < 0.05.

* Represents OR for every 10 mm Hg increase in PaO₂. Controlling for median post-cannulation PaCO₂.

† Represents OR for every 10 mm Hg increase in PaCO₂. Controlling for median post-cannulation PaO₂.

Hypoxemia post-cannulation is defined as a median PaO₂ value below 70 mm Hg.

ΔPaO₂, peri-cannulation arterial oxygen drop; ΔPaCO₂, peri-cannulation arterial carbon dioxide drop; CI, confidence interval; OR, odds ratio; PaO₂ arterial oxygen tension; PaCO₂, arterial carbon dioxide tension.